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Roll No

FT-4004-CBGS

B.E. IV Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Fluid Mechanics

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Differentiate between :
 - i) Liquids and Gases
 - ii) Real fluids and Ideal fluids
 - iii) Specific weight and specific volume of a fluid
- b) A rectangular plane surface is 2m wide and 3m deep. It lies in vertical plane in water. Determine the total pressure and position of centre of pressure on the plane surface when its upper edge is horizontal and coincide with water surface.
2. a) Define the terms (i) Velocity potential function (ii) Stream function. Prove that equipotential line and line of constant stream function are orthogonal to each other.
- b) Check if $\phi = x^2 - y^2 + y$ represent the velocity potential for 2-dimensional irrotational flow. If it does then determine the stream function ψ .
3. a) Derive an expression for Bernoulli's theorem from first principle and state the assumptions made.

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- b) A pipe line carrying oil of specific gravity 0.87, changes in diameter from 200 mm at position A to 500 mm diameter at position B which is 4 m at higher level. If the pressure at A is 9.81 N/cm^2 and at B 5.886 N/cm^2 and discharge is 200 litre/sec. Determine the loss of head and direction of flow.
4. Water is flowing through a pipe having diameters 30 cm and 15cm at the bottom end and upper end respectively. The intensity of pressure at bottom end is 29.43 N/cm^2 and at the upper end is 14.715 N/cm^2 . Determine the difference in datum head if the rate of flow through the pipe is 50 lit/s.
5. a) Explain the principle of venturimeter with a neat sketch. Derive the expression for the rate of flow through venturimeter.
b) A 30 cm×15 cm venturimeter is inserted in a vertical pipe carrying water, flowing in the upward direction. A differential mercury manometer connected to the inlet and throat gives a reading of 30 cm. Find the discharge. Take $C_d = 0.98$.
6. a) State Buckingham's π theorem. Why this theorem is considered superior over the Rayleigh's method analysis.
b) Starting with Euler's equation of motion along a streamline and integrate it to obtain Bernoulli's equation. State all assumptions made.
7. What is a venturimeter? Derive an expression for the discharge through a venturimeter.
8. Write short notes on any two of following.
a) Stability of submerged bodies
b) Rotational and irrotational flow
c) Boundary layer separation

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